Wenbo Zhang

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Research Interests

My research explores **human-centered wearable and ubiquitous intelligence**, with broad interests in multimodal AI, human sensing, mobile and ubiquitous systems, and smart health. I develop sensing and learning methods that enable computing systems to understand human states, behaviors, and contexts from heterogeneous real-world data, while remaining adaptive, generalizable, and personalized across users, devices, and environments. More broadly, I am interested in human–computer interaction, assistive technologies, and intelligent systems that improve health, well-being, and everyday human experiences.

Education

South China University of Technology, Guangzhou, China — *Sep 2023 – Present*
Ph.D. Candidate in Information and Communication Engineering (Expected Jun 2027)
Supervisor: Prof. Zhanpeng Jin

University College London (UCL), London, UK — *Mar 2026 – Aug 2026*
Visiting Researcher, Department of Computer Science
Host: Prof. Jagmohan Chauhan

Institute of Computing Technology, Chinese Academy of Sciences, Beijing, China — *May 2021 – Jun 2023*
Joint M.S. Research Program in Computer Science

Hebei University, Baoding, China — *Sep 2020 – Jun 2023*
M.S. in Computer Science

Hebei University of Science and Technology, Shijiazhuang, China — *Sep 2016 – Jun 2020*
B.S. in Internet of Things Engineering

Publications

[P1] Wenbo Zhang, Xingying Yan, Chenxu Zhang, Dong Huitian, Haoyang Li, Chenxu Zhu, Yang Gao, Jagmohan Chauhan, and Zhanpeng Jin. "**HALO**: Inferring Hidden Activation–Loading Organization from Sparse Wearables." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 2026. **Accepted.**

[P2] Wenbo Zhang, Yuan Gao, Huitian Dong, Chenxu Zhu, Chenxu Zhang, Wenkang Zhang, Yang Gao, Jagmohan Chauhan, and Zhanpeng Jin. "Beyond Sensor Streams: **HALO-Agent** as a Body-State Representation Layer for Wearable Agents." *WearAgent Workshop, ACM UbiComp/ISWC 2026*. **Accepted.**

[P3] Wenbo Zhang, Chenxu Zhang, Xingying Yan, Guanyu Xin, Yu He, Wenkang Zhang, Jagmohan Chauhan, Yang Gao, and Zhanpeng Jin. "**MuscleSense**: Making Muscle Effort Audible for Eyes-Free In-the-Loop Regulation." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 10(3), Article 190, 2026.

[P4] Wenbo Zhang, Chenxu Zhang, Yang Gao, and Zhanpeng Jin. "**KineticsSense**: A Multimodal Wearable Sensor Framework for Modeling Lower-Limb Motion Kinetics." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 9(3), Article 151, 2025.

[P5] Lingde Hu[†], **Wenbo Zhang**[†], Seokmin Choi, Yang Gao, Jagmohan Chauhan, and Zhanpeng Jin. "**WatchForce**: Wrist-worn PPG and IMU Sensing for Hand Force Estimation." *Computer Methods and Programs in Biomedicine*, vol. 285, Article 109547, 2026. ([†] Equal contribution.)

[P6] Lingde Hu[†], **Wenbo Zhang**[†], Wenkang Zhang, Yu He, Seokmin Choi, Yang Gao, Jagmohan Chauhan, and Zhanpeng Jin. "**PPGSpeech**: A Wearable Silent Speech Interface Leveraging Neck-Worn Photoplethysmography." *IEEE Internet of Things Journal*, 13(4), pp. 6692–6703, 2026. ([†] Equal contribution.)

[P7] Yang Gao, **Wenbo Zhang**, Junbin Ren, Ruihao Zheng, Yingcheng Jin, Di Wu, Lin Shu, Xiangmin Xu, and Zhanpeng Jin. "**PressInPose**: Integrating Pressure and Inertial Sensors for Full-Body Pose Estimation in Activities." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 8(4), Article 197, 2024.

[P8] Yang Gao, Yingjing Xiao, Junbin Ren, Chenxu Zhang, **Wenbo Zhang**, and Zhanpeng Jin. "Trading Sensing for Structure: Sparse IMU–EMG Fingertip Force Estimation via Neuro-inspired Structured Modeling." *Advances in Neural Information Processing Systems (NeurIPS)*, 2026. **Accepted.**

[P9] Junbin Ren, Ruihao Zheng, **Wenbo Zhang**, Dong She, Yuting Bai, Zhanpeng Jin, and Yang Gao. "**Motion2Press**: Cross-Modal Learning from IMU to Plantar Pressure for Gait Analysis." *Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. (IMWUT)*, 9(3), Article 124, 2025.

[P10] Zhenchao Cui, **Wenbo Zhang**, Zhaoxin Li, and Zhaoqi Wang. "Spatial–Temporal Transformer for End-to-End Sign Language Recognition." *Complex & Intelligent Systems*, 9(4), pp. 4645–4656, 2023.

Manuscripts Under Review

[M1] Wenbo Zhang et al. "**BEAM**: Biomechanical Evidence-Aligned Movement Reasoning from Sparse Wearable Sensing." *IEEE Internet of Things Journal*, 2026. **(Under Review)**

[M2] He Yu, Wenkang Zhang, **Wenbo Zhang**, Lingde Hu, Muhammad Adil, Yincheng Jin, and Zhanpeng Jin. "A Survey on Brain Decoding: Reconstruction of Audio-Visual Perception." *ACM Computing Surveys*, 2026. **(Third-round Revision Under Review)**

One additional manuscript is under double-blind review; identifying details are withheld to preserve submission anonymity.

Patents

[PT1] Zhanpeng Jin, **Wenbo Zhang**, and Yang Gao. "A Full-Body Pose Estimation Method and System Based on Pressure and Inertial Sensors." China Patent **ZL 2024 1 1590666.7**, granted Aug. 8, 2025 (Publication No. CN 119645223 B).

Research Agenda

- **Multimodal Human Sensing and Computational Modeling.** I seek to develop wearable and mobile sensing systems that capture and model human movement, physiological responses, behavior, and latent biomechanical states in everyday environments. A central goal is to move beyond surface-level activity recognition toward richer representations of human state that can support movement understanding, health monitoring, rehabilitation, and human performance. Representative work includes **HALO [P1]**, **KineticsSense [P4]**, **PressInPose [P7]**, and **Motion2Press [P9]**.
- **Adaptive and Generalizable AI for Human-Centered Sensing.** I aim to build learning methods that remain reliable when sensing modalities, devices, users, activities, and deployment conditions change. My interests include multimodal and cross-modal representation learning, transfer learning, personalization, continual learning, uncertainty estimation, and learning from sparse or weakly available sensing. Current work such as **BEAM [M1]** and **StateScope** extends this direction toward transferable representations and principled characterization of what human states can be reliably inferred across sensing conditions, building on **HALO [P1]** and **Trading Sensing for Structure [P8]**.
- **Human-Centered Intelligent Systems for Interaction and Health.** I explore how inferred human states can be translated into real-time feedback, adaptive interaction, personalized assistance, and longitudinal intelligent systems. I am particularly interested in closing the loop between sensing, reasoning, and intervention for smart health, physical activity, rehabilitation, and natural human–computer interaction. Representative work includes **MuscleSense [P3]**, **WatchForce [P5]**, **PPGSpeech [P6]**, and **HALO-Agent [P2]**. Ongoing work extends toward effort-aware interaction, longitudinal personalization, and evidence-grounded wearable agents.

Selected Talks and Presentations

- **ACM UbiComp/ISWC 2026**, Shanghai, China, Oct. 14, 2026. "MuscleSense: Making Muscle Effort Audible for Eyes-Free In-the-Loop Regulation." *Scheduled oral presentation.*
- **WearAgent 2026 Workshop, ACM UbiComp/ISWC 2026**, Shanghai, China, Oct. 12, 2026. "Beyond Sensor Streams: HALO-Agent as a Body-State Representation Layer for Wearable Agents." *Scheduled workshop presentation.*
- **HHME 2026**, Changsha, China, Aug. 17, 2026. "KineticsSense: A Multimodal Wearable Sensor Framework for Modeling Lower-Limb Motion Kinetics." *Research presentation.*
- **UCL Interaction Centre (UCLIC)**, University College London, London, UK, July 22, 2026. "Beyond Motion: Body-State-Aware Wearable Intelligence for Human Movement and Interaction." *Research seminar.*
- **MobiUK 2026**, University of Cambridge, UK, June 26, 2026. "MuscleSense: Real-Time Auditory Biofeedback for Lower-Limb Exercise."
- **ACM UbiComp/ISWC 2025**, Espoo, Finland, Oct. 16, 2025. "KineticsSense: A Multimodal Wearable Sensor Framework for Modeling Lower-Limb Motion Kinetics." *Oral presentation.*

Mentoring and Teaching

Research Mentoring, South China University of Technology *2023 – Present*

- Mentored **14 graduate and undergraduate researchers** across wearable sensing, multimodal learning, human-subject experimentation, machine learning, and scientific writing, supervising research from problem formulation and data collection to modeling and manuscript preparation.
- **Selected mentoring outcomes:** Xiaozhe Liang secured admission to the **Ph.D. program at Nanjing University** (for 2027); Xingying Yan secured admission to the **M.S. program at Zhejiang University** (for 2027); and Chenxu Zhang, whom I have mentored since his undergraduate studies, is now a **first-year M.S. student at SCUT** and continues our research collaboration.
- **Current and recent mentees:** Chenxu Zhu (Ph.D. student, 2025 cohort); Chenxu Zhang (M.S. student, SCUT); Xiaozhe Liang, Xingying Yan, Huitian Dong, and Haoyang Li (undergraduate, 2023 cohort); Yuan Gao, Shijia Jiao, Guanyu Xin, Ruihan Weng, and Zhan Wang (undergraduate, 2024 cohort); Haoxuan Qi and Boyi Wang (undergraduate, 2025 cohort); and Samadhi Herath (MSci student, University College London), with cross-institutional research mentoring on machine learning and multimodal human-movement sensing.

Teaching Assistant, Introduction to Engineering *2024 – 2026*

- Mentored two six-student first-year engineering design teams in consecutive cohorts; both teams received Second Prize in the annual course design competition.

Academic Service

- **Journal Reviewer:** *Pervasive and Mobile Computing* (Elsevier), 2026.
- **Conference Reviewer:** ACM UbiComp/ISWC, ACM ISS, and AAAI, 2026.
- **Invited Panelist:** "Impact, Risks, and Security/Privacy in Ubiquitous Computing Research," ACM UbiComp/ISWC 2025.
- **Workshop Organization:** Contributed to organizing WearAgent 2026, ACM UbiComp/ISWC 2026.
- **Student Volunteer:** ACM UbiComp/ISWC 2025, Espoo, Finland.

Selected Honors

- China Scholarship Council Joint Ph.D. Scholarship, 2025.
- Silver Medal, Huawei Ascend AI Innovation Competition, 2023.
- Research Excellence Award for Graduate Students, 2023.